

System Architecture & Technical Blueprint

Product Name: HQ

Tagline: Your AI Headquarters

Version: 1.0

Purpose

This document defines the complete technical architecture of HQ.

Our goals are:

Enterprise-grade architecture

AI-first orchestration

High scalability

Maintainability

Fast AI-assisted development

Production readiness

Architecture Philosophy

HQ will use a modular monolith for the MVP.

Why not microservices?

Because for a hackathon and early startup:

Faster development

Simpler deployment

Easier debugging

Lower infrastructure cost

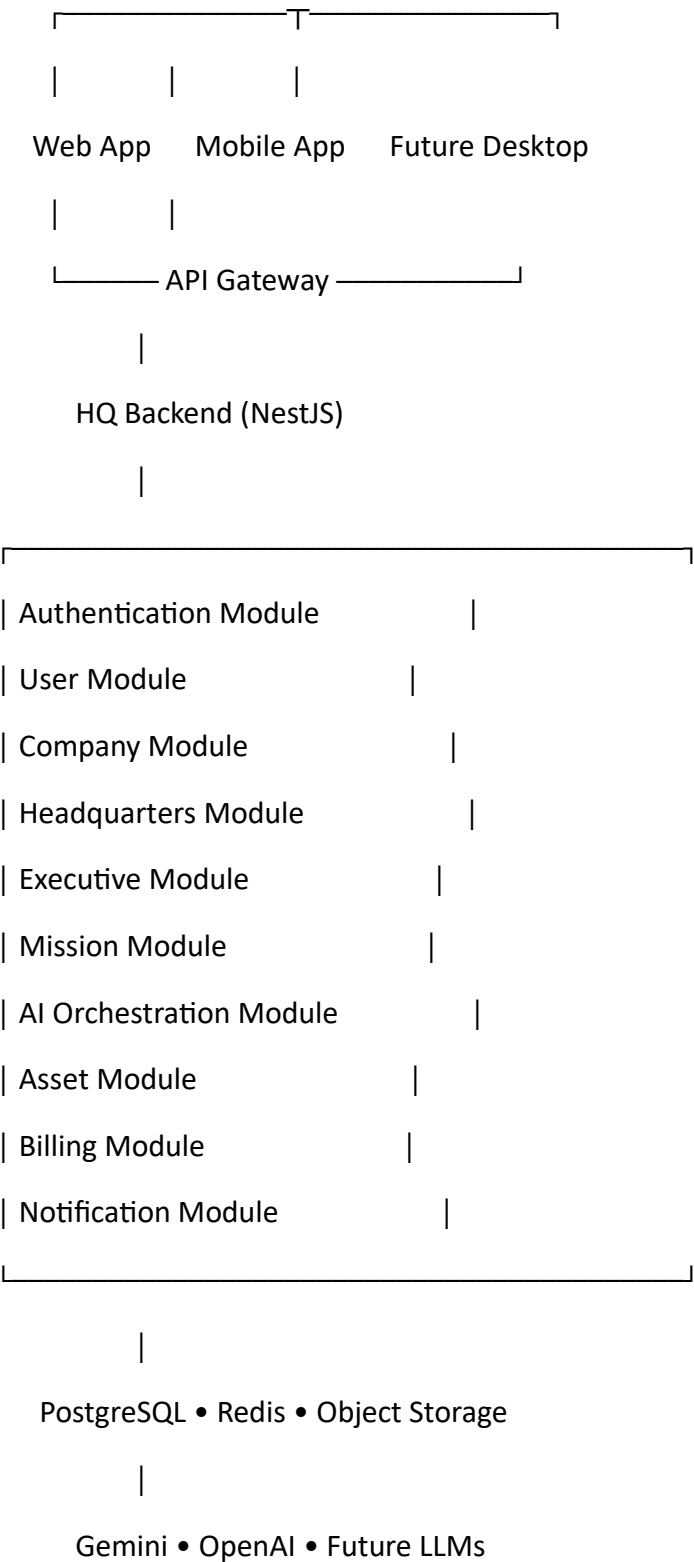
Easier AI code generation

The architecture will be designed so modules can later be extracted into microservices if needed.

High-Level Architecture

Plain text

Client Applications



Monorepo Structure

Plain text

hq/

|

└─ apps/

| └─ web/

| └─ api/

| └─ mobile/

|

└─ packages/

| └─ ui/

| └─ types/

| └─ config/

| └─ ai/

| └─ shared/

|

└─ docs/

└─ scripts/

└─ docker/

└─ tools/

Backend Modules

Core

Authentication

Users

Companies

Brands

Headquarters

Departments

Executives

Missions

Tasks

AI

CEO Engine

Chief of Staff Engine

Executive Engine

Prompt Engine

Memory Engine

AI Provider Manager

Business

Assets

Billing

Notifications

Analytics

Settings

AI Orchestration Engine

This is the heart of HQ.

Instead of directly calling an AI model:

Plain text

User

↓

CEO



Chief of Staff



Executive Board



Mission Plan



Task Execution



Review



Final Delivery

Every mission flows through the orchestration engine.

Event-Driven Architecture

Modules communicate through internal events such as:

MissionCreated

MissionAssigned

ExecutiveStarted

ExecutiveCompleted

AssetGenerated

MissionCompleted

PaymentSucceeded

SubscriptionChanged

This reduces coupling between modules.

Background Jobs

Long-running tasks will be processed asynchronously.

Examples:

AI generation

Image generation

Email delivery

Notifications

Analytics updates

Real-Time Updates

WebSockets will power:

Executive status

Mission progress

Notifications

Live collaboration timeline

This makes HQ feel alive.

Storage Strategy

PostgreSQL → Structured data

Redis → Cache and queues

Object Storage → Images, videos, documents

Vector Database (future) → Semantic memory and retrieval

Security

JWT authentication

Refresh tokens

Role-based access control

API rate limiting

Input validation

Encryption for sensitive data

Audit logs

External Integrations

The architecture will support:

Gemini

OpenAI

Stripe

Email provider

Cloud storage

Social media APIs (future)

Scalability

Designed to support:

Multiple organizations

Multiple Headquarters

Multiple AI providers

Multiple workspaces

Thousands of concurrent users

without redesigning the core architecture.

Development Principles

Every module must be:

Independent

Testable

Reusable

Well documented

AI-friendly

Replaceable

Scalable

Technical Vision

HQ is not being built as a hackathon prototype.

It is being built as the foundation of a production-ready AI business operating platform capable of evolving into a global SaaS product.